

Student Name: Jinpeng Zhu
Student ID: 1907843
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Intro to Finite Element Method (CIV E – 665)
Assignment No. 2



1 Question 1

1.1 i

The given strong form:

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) - b(x) = 0 \quad (1)$$

Since $b(x) = 0$, the strong form becomes:

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = 0 \quad (2)$$

Therefore, for this question, the weighted integral statement is,

$$\int_0^L w(x) \left[-\frac{d}{dx} \left(EA \frac{du}{dx} \right) \right] dx = 0 \quad (3)$$

1.2 ii

Apply the integration by parts rule to the Equation 3,

$$-\left[wEA \frac{du}{dx} \right]_0^L + \int_0^L \frac{dw}{dx} EA \frac{du}{dx} dx = 0 \quad (4)$$

Expand the bracket,

$$-w(L)EA \frac{du}{dx} \Big|_L + w(0)EA \frac{du}{dx} \Big|_0 + \int_0^L EA \frac{dw}{dx} \frac{du}{dx} dx = 0 \quad (5)$$

We try to end up with the same continuity requirement between $u(x)$ and $w(x)$

- $u(x)$: must at least be twice differentiable (because we have $\frac{d}{dx} \left(EA \frac{du}{dx} \right)$)
- $w(x)$: No differentiability requirement (because we have $w(x)$)

Boundary term for this problem,

- Case 1: Dis appears when we have an essential B.C, at $x = 0, u(0) = 0$, so we impose $w(0) = 0$
- Case 2: At $x = L, EA \frac{du}{dx} = 2EA_0 \frac{du}{dx} = F$

Substituting these into Equation 5, the final weak form is:

$$-w(L)F + \int_0^L EA \frac{dw}{dx} \frac{du}{dx} dx = 0 \quad (6)$$

1.3 iii

Given $U_1(x) = c_1x + c_0$, apply B.C case 1, $c_0 = 0$ can be obtained, so

$$\frac{dU_1}{dx} = c_1 \quad (7)$$

Per Ritz/Galerkin Method, the weight function has the same mathematical form as the approximate solution; therefore, set:

$$w(x) = x \quad \Rightarrow \quad \frac{dw}{dx} = 1 \quad (8)$$

Substituting known data: $E = 100 \text{ GPa} = 100 \times 10^9 \text{ Pa}$, $A_0 = 0.005 \text{ m}^2$, $F = 100 \text{ N}$, $L = 1 \text{ m}$, $EA_0 = (100 \times 10^9)(0.005) = 5 \times 10^8 \text{ N}$, $A(x) = A_0(1+x)$ and Equation 7, Equation 8 into Equation 6,

$$\begin{aligned} -w(1)F + \int_0^1 EA_0(1+x)(1)c_1 dx &= 0 \\ \Rightarrow -100 + 5 \times 10^8 \int_0^1 (1+x)(1)c_1 dx &= 0 \\ \Rightarrow -100 + 5 \times 10^8 c_1 \left[x + \frac{x^2}{2} \right]_0^1 &= 0 \\ \Rightarrow -100 + (5 \times 10^8)c_1(1+0.5) &= 0 \\ \Rightarrow c_1 &\approx 1.333 \times 10^{-7} \end{aligned}$$

So,

$$U_1(x) = \frac{100}{5 \times 10^8 \times 1.5} x = 1.333 \times 10^{-7} x$$

1.4 iv

For using an =2 Ritz/Galerkin Approximation,

$$U_2(x) = c_1x + c_2x^2 + c_0 \quad (9)$$

Apply essential B.C. $u(0) = 0 \Rightarrow c_0 = 0$,

$$\begin{aligned} U_2(x) &= c_1x + c_2x^2 \\ \frac{dU_2}{dx} &= \sum c_j \frac{d\theta_j}{dx} \end{aligned} \quad (10)$$

Based on the given data, we know $\theta_1 = x$, $\theta_2 = x^2$

To keep the same mathematical form as the $U_2(x)$,

$$w(x) = x + x^2$$

Re-arrange Equation 6 into $\{K\}\{c\} = \{f\}$,

$$\int_0^L EA \frac{dw}{dx} \frac{du}{dx} dx = w(L)F$$

Building Ritz/Galerkin system, we use index i to evaluate all positions j

$$\int_0^L EA \frac{d\theta_i}{dx} \left(\sum_{j=1}^n c_j \frac{d\theta_j}{dx} \right) dx = \theta_i(L)F \quad (11)$$

Discretize Equation 11

$$\sum_{j=1}^n \overbrace{\left[\int_0^L EA(x) \frac{d\theta_i}{dx} \frac{d\theta_j}{dx} dx \right]}^{K_{ij}} \frac{c_j}{\hat{c}_j} = \overbrace{\theta_i(L)F}^{f_i} \quad (12)$$

Then we can get 4 equations to form a stiffness matrix K:

$$\begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \quad (13)$$

Substituting known data, we can get,

$$\begin{bmatrix} \int_0^1 EA_0(1+x)(1)(1)dx & \int_0^1 EA_0(1+x)(1)(2x)dx \\ \int_0^1 EA_0(1+x)(2x)(1)dx & \int_0^1 EA_0(1+x)(2x)(2x)dx \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 100 \end{bmatrix} \quad (14)$$

To here, we can easily get unknown c_1 and c_2 by any method. Now we use Python to assist in solving Equation 14. The code is shown below:

```
import sympy as sp

# Define the symbolic variable
x = sp.symbols('x')

# Constants
E = 100 * 10**9
A0 = 0.005
L = 1
F = 100
EAO = E * A0 # This is a constant coefficient (5 * 10^8)

# Define the four integral terms in the stiffness matrix K
K11 = sp.integrate(EAO * (1 + x) * 1 * 1, (x, 0, 1))
K12 = sp.integrate(EAO * (1 + x) * 1 * (2*x), (x, 0, 1))
K21 = sp.integrate(EAO * (1 + x) * (2*x) * 1, (x, 0, 1))
K22 = sp.integrate(EAO * (1 + x) * (2*x) * (2*x), (x, 0, 1))

# 4. Build the K matrix and force vector
K_matrix = sp.Matrix([[K11, K12], [K21, K22]])
F_vector = sp.Matrix([100, 100])

# Solve [K]{c} = {f}
solution = K_matrix.LUsolve(F_vector) # Using LU decomposition
c1, c2 = solution[0], solution[1]
```

c_1	c_2
$1.84615384615385 \cdot 10^{-7}$	$-4.61538461538461 \cdot 10^{-8}$

Therefore, the final approximate solution is,

$$U_2(x) = 1.84615384615385E - 7x + -4.61538461538461E - 8x^2$$

1.5 v

Using Matplotlib to plot $u(x)$, $U_1(x)$ and $U_2(x)$ together, as shown in the Figure 1. The code is shown below:

```
import numpy as np
import matplotlib.pyplot as plt
# Generate x values
x_vals = np.linspace(0, 1, 200)

# Define the exact solution function
u = F*L / (E * A0) * np.log(1 + x_vals/L)
U1 = (100/(5 * 10**8 * 1.5)) * x_vals
U2 = c1 * x_vals + c2 * x_vals**2
# Calculate corresponding u(x), U1(x), and U2(x) values
u_vals = F*L / (E * A0) * np.log(1 + x_vals/L)
U1_vals = (100/(5 * 10**8 * 1.5)) * x_vals
U2_vals = c1 * x_vals + c2 * x_vals**2

# Plotting
plt.figure(figsize=(8, 6))
plt.plot(x_vals, u_vals, label='Exact $u(x)$')
plt.plot(x_vals, U1_vals, label='$U_1(x)$')
plt.plot(x_vals, U2_vals, label='$U_2(x)$')
plt.xlabel('x')
plt.ylabel('Displacement')
plt.title('Comparison of Exact and Approximate Solutions')
plt.legend()
plt.grid(True)
plt.show()
```

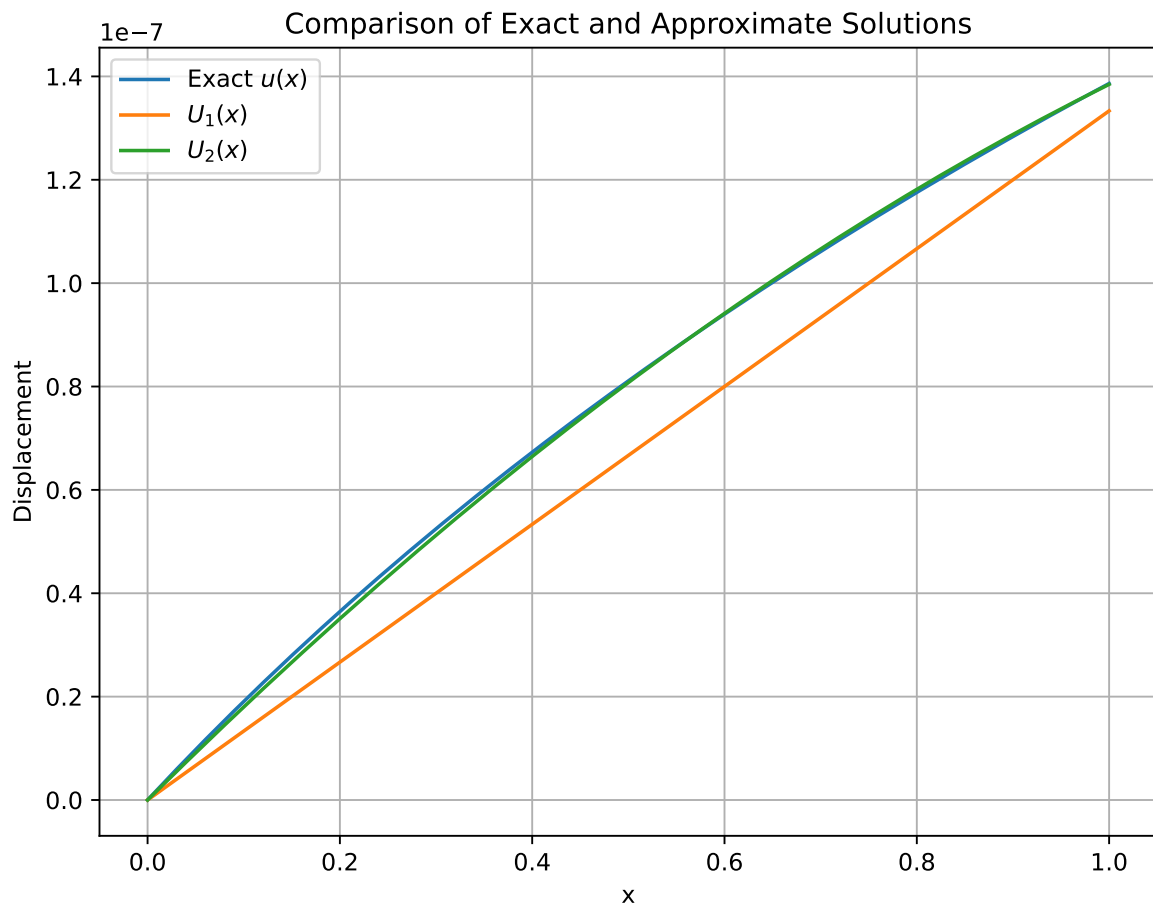


Figure 1: Comparison of Exact and Approximate Solutions

2 Question 2

My answer: I successfully installed Abaqus.